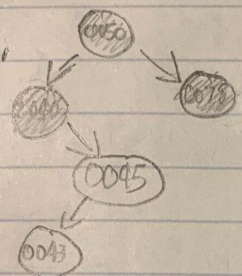
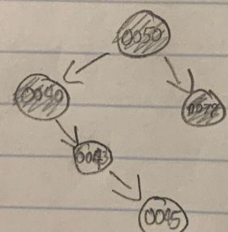
Octavio Morales
Homework 5
I pledge my honor that I have abided by the Stevens Honor System. Ocm

1. a.



-black
O-red

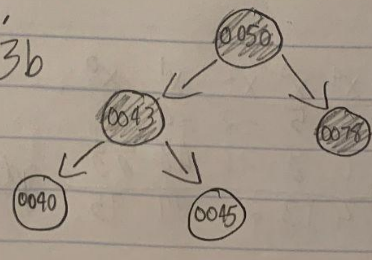
b. Property 4 is violated because there is a red node child for a red node.
Case Seen: 2b



-black
O-red

Steps: z=parent[z], rightRotate[z]

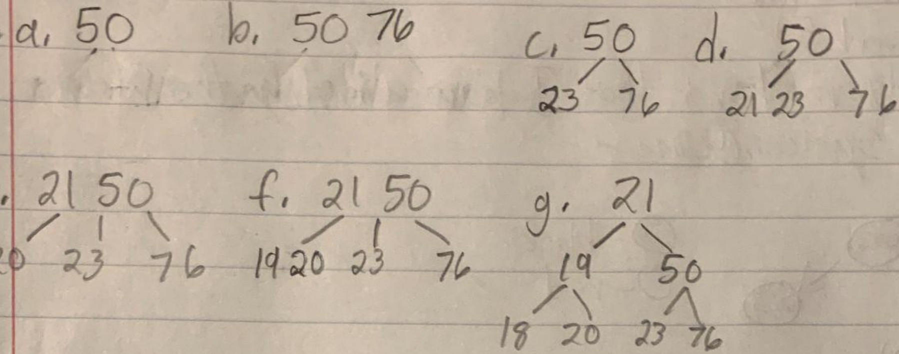
c. Property 4 is violated because there is a red node child for a red node.
Case Seen: 3b



-black
O-red

Steps: parent[z].color=black, parent[parent[z]].color=red, leftRotate(parent[parent[z]])

2\. a. b. c. d. e. f. g.



3\. LCM(A[1...n]):

```
answer = A[0]
for loop (x=1 to x=n-1)
    answer = answer · A[x]/gcd(answer, A[x])
return answer
```

4\. a. $p(x) = 4x^4 + 5x^3 - 2x^2 - 4x + 7$

$p(x) = x(4x^3 + 5x^2 - 2x - 4) + 7$

$p(x) = x(x(4x^2 + 5x - 2) - 4) + 7$

$p(x) = x(x(x(4x + 5) - 2) - 4) + 7$

$p(x) = x(x(x(x(4) + 5) - 2) - 4) + 7$

b. $p = [7, -4, -2, 5, 4]$

c.

x	p	n	i
2	4	4	
	13		3
	24		2
	44		1
	95		0

$p(2) = 95$

d.

x_0	x^4	x^3	x^2	x^1	x^0
2	4	5	-2	-4	7
	0	8	26	48	88
0	4	13	24	44	95

$p(2) = 95$

5. LeftRightBinaryExponentiation(a, b(n)):

```
answer=1
for loop(x to 0)
    answer=answer*answer
    if(b_x==1)
        answer=answer*a
return answer
```